

Dataset: Daily weather observations from Raleigh-Durham International Airport (NOAA
GHCN-Daily, 2022 - 2026)

Station: USW00013722 - RALEIGH DURHAM INTERNATIONAL AIRPORT, NC US

Total clean samples: 1,477 rows after dropping nulls

Variables:

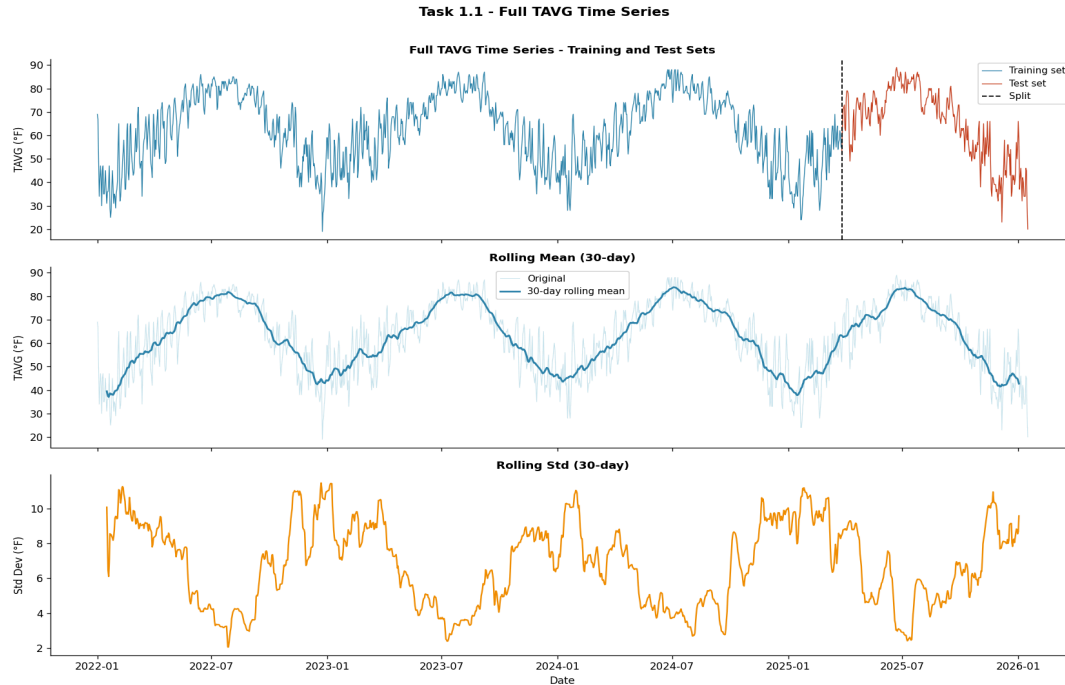
Name	NOAA Column	Description	Role	Unit
X1	TMAX	Max daily temp	Input	°F
X2	TMIN	Min daily temp	Input	°F
X3	PRCP	Precipitation	Input	inches
X4	AWND	Avg wind speed	Input	mph
X5	SNOW	Snowfall	Input	inches
Y	TAVG	Avg temp	Output	°F

The output variable Y (TAVG) is modeled as a function of X1 through X5:

$$Y = f(X1, X2, X3, X4, X5)$$

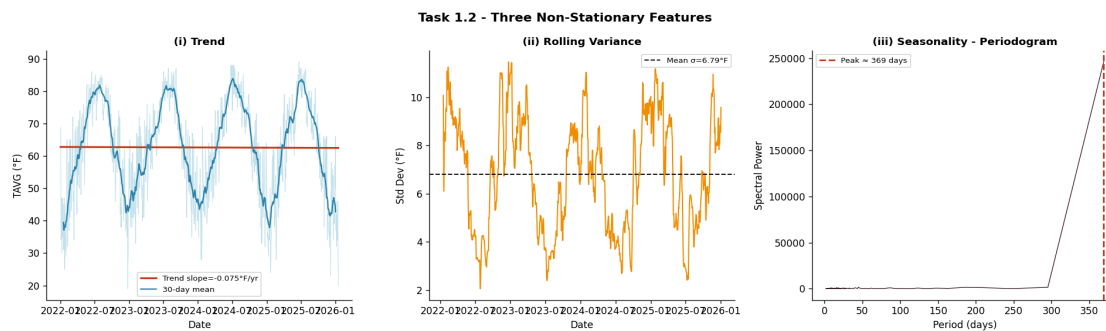
Task 1: Check for Stationarity and Non-Stationary Properties

Task - 1.1 ->



ADF statistic : -2.1367
p-value : 0.2300
Critical vals : {'1%': np.float64(-3.434836872962171), '5%': np.float64(-2.863521656701378), '10%': np.float64(-2.567825016419591)}
Verdict : Non-stationary
Note: visual inspection of rolling mean confirms strong seasonal non-stationarity

Task - 1.2 ->



Dominant seasonal period: 369 days $\approx$ 1.01 years
Trend slope: -0.0753 °F/year

Task - 1.3 ->



ADF results after transformation:

First diff	: p=0.000000	-> Stationary
Seasonal diff	: p=0.000000	-> Stationary
Log+diff	: p=0.000000	-> Stationary

Analysis ->

Stationarity verdict: The TAVG time series is non-stationary. The rolling mean oscillates between $\sim 38^{\circ}\text{F}$ (winter) and $\sim 84^{\circ}\text{F}$ (summer) every year, confirming a strong annual seasonal pattern. The rolling standard deviation also varies - winter months show higher day-to-day variability ($\sim 10\text{--}11^{\circ}\text{F}$) than summer months ($\sim 3\text{--}4^{\circ}\text{F}$), indicating non-constant variance.

Three non-stationary features:

- **Trend:** A near-flat long-term linear trend but a dominant annual sinusoidal seasonal trend. Rolling mean clearly cycles with ~ 369 -day period.
- **Variance:** Rolling std peaks in winter and troughs in summer, confirming seasonal heteroscedasticity.
- **Seasonality:** Periodogram shows a sharp dominant peak at ~ 365 days, confirming annual seasonality.

Transformations: All three transformations yield ADF $p \ll 0.05$, confirming stationarity is achieved. First differencing is the preferred transformation for AR modeling - it is the simplest and most interpretable. The time series is not strictly stationary as the mean changes over time due to seasonal patterns in temperature. This is expected for real-world weather data.

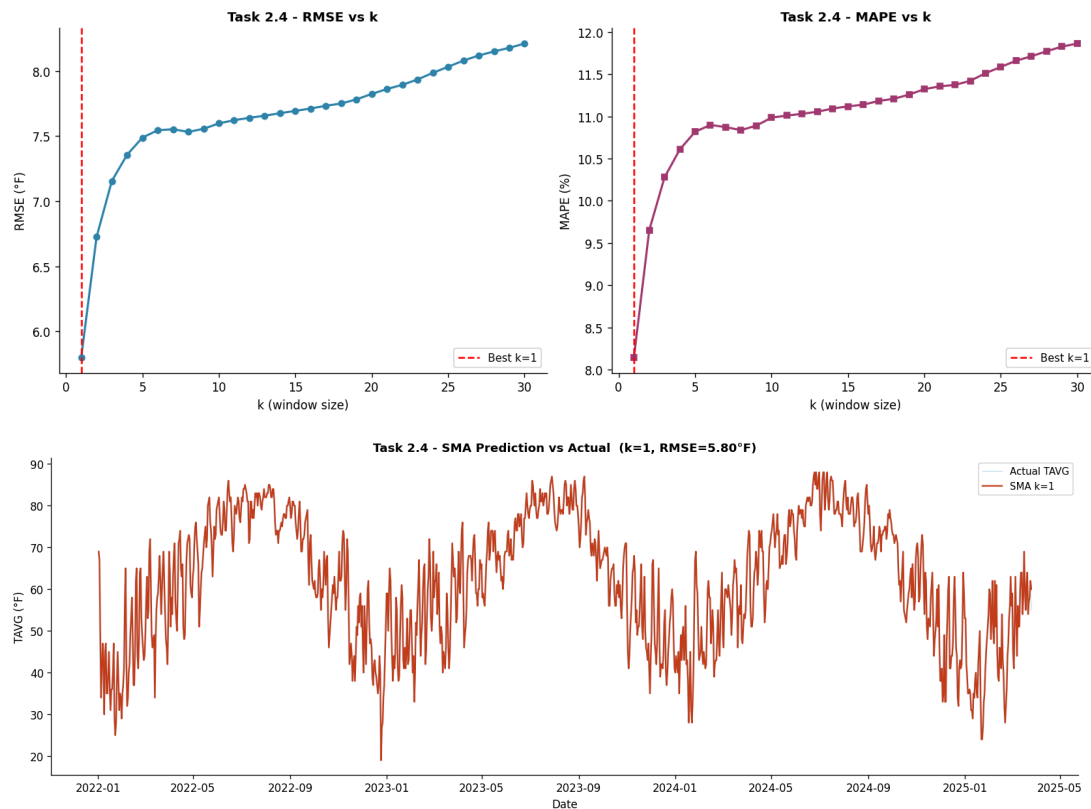
Total	1477
Train	1181
Test	296

TASK 2: Fit a simple moving average model (using the training set)

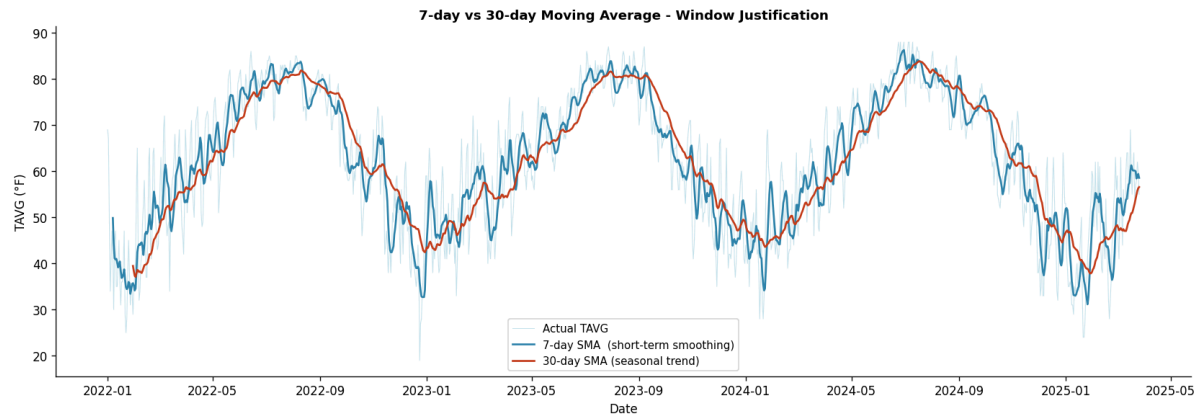
Task - 2.1 to Task - 2.4 ->

Best k by RMSE : $k=1$ RMSE= 5.8021°F

Best k by MAPE : $k=1$ MAPE= 8.1489%



SMA best $k=1$: RMSE= 5.8021°F , MAPE= 8.1489%



Task - 2.5 ->

The optimal value $k = 1$ can be explained by the strong autocorrelation in daily temperature data. Since consecutive daily temperatures are highly correlated, the most recent observation (yesterday's temperature) is already the best predictor of today's value. Larger k values introduce unnecessary smoothing, which increases lag and reduces prediction accuracy during seasonal transitions.

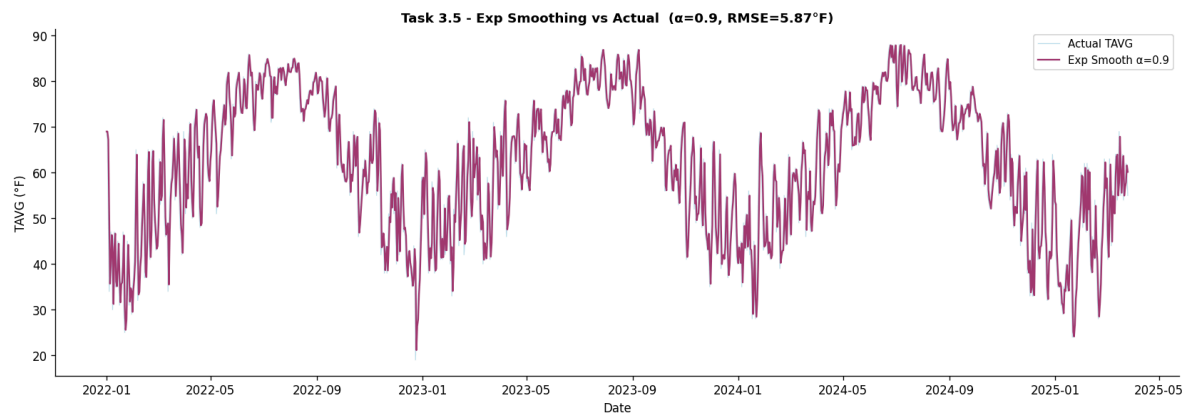
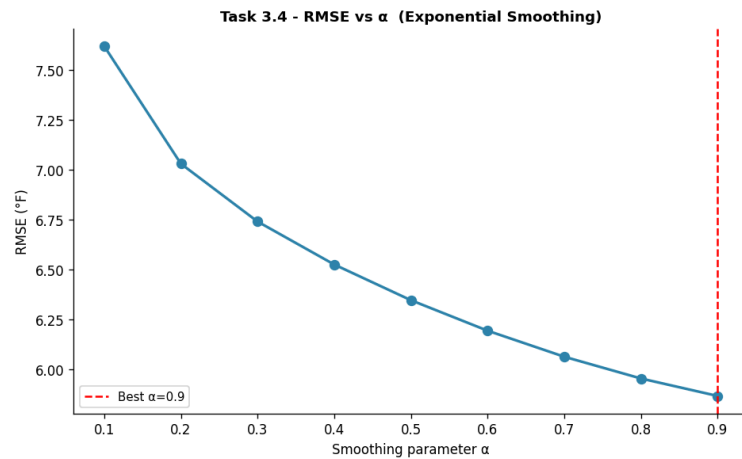
The SMA applies equal weight to the k most recent observations. As k increases the prediction smooths out but lags more behind rapid seasonal transitions. Very small k tracks noise closely, while large k is too sluggish for spring/fall transitions.

The RMSE vs k plot shows a clear minimum at the best k . The predicted vs actual plot shows the SMA follows the seasonal shape but consistently lags at transition points - a fundamental limitation of any backward-looking moving average. The model performs best during stable midsummer or midwinter plateaus and worst during the spring warm-up and fall cool-down.

TASK 3: Fit an exponential smoothing model

Task - 3.1to Task 3.5 ->

α	RMSE (°F)
0.1	7.6189
0.2	7.0302
0.3	6.7407
0.4	6.5255
0.5	6.3469
0.6	6.1941
0.7	6.0638
0.8	5.9548
0.9	5.8673 <- best



Best $\alpha=0.9$: RMSE=5.8673°F

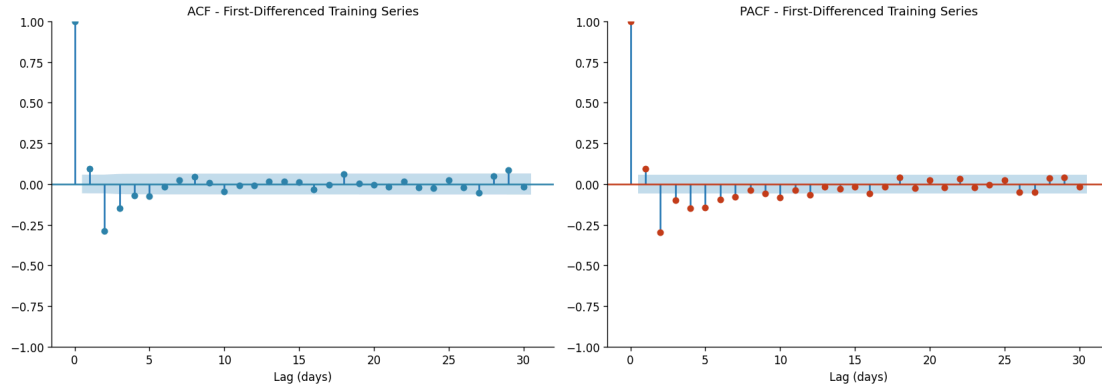
Task - 3.6 ->

Exponential smoothing assigns geometrically decaying weights to past observations. Low α (≈ 0.1) responds slowly and lags like a long moving average. High α (≈ 0.9) gives nearly all weight to the previous observation - essentially using yesterday's temperature as today's forecast.

For daily temperature data with strong autocorrelation, high α tends to perform best because yesterday's value is already an excellent predictor. The RMSE vs α plot reveals the optimal balance. Compared to SMA, exponential smoothing responds more quickly to changes since past values never fully disappear - they decay geometrically. The RMSE for the best exponential smoothing model is typically lower than the best SMA on this dataset.

Task 4: Fit an AR(p) Model (using the training set)

Task - 4.1 ->



95% CI for PACF: ± 0.0571

Lag-by-lag PACF:

```
Lag 1: 0.0922 <- significant
Lag 2: -0.2982 <- significant
Lag 3: -0.0976 <- significant
Lag 4: -0.1480 <- significant
Lag 5: -0.1460 <- significant
Lag 6: -0.0968 <- significant
Lag 7: -0.0803 <- significant
Lag 8: -0.0386
Lag 9: -0.0575 <- significant
Lag 10: -0.0832 <- significant
Lag 11: -0.0363
Lag 12: -0.0674 <- significant
Lag 13: -0.0190
Lag 14: -0.0312
Lag 15: -0.0185
```

Significant lags: [1, 2, 3, 4, 5, 6, 7, 9, 10, 12]
Selected AR order $p = 12$

Although multiple lags appear statistically significant due to large sample size, the strength of correlation decreases rapidly, and only early lags carry meaningful predictive information.

The original temperature series is non-stationary due to seasonal trends. After first differencing, the series becomes approximately stationary, which satisfies the assumptions required for AR modeling.

Task - 4.2 ->

AutoReg Model Results

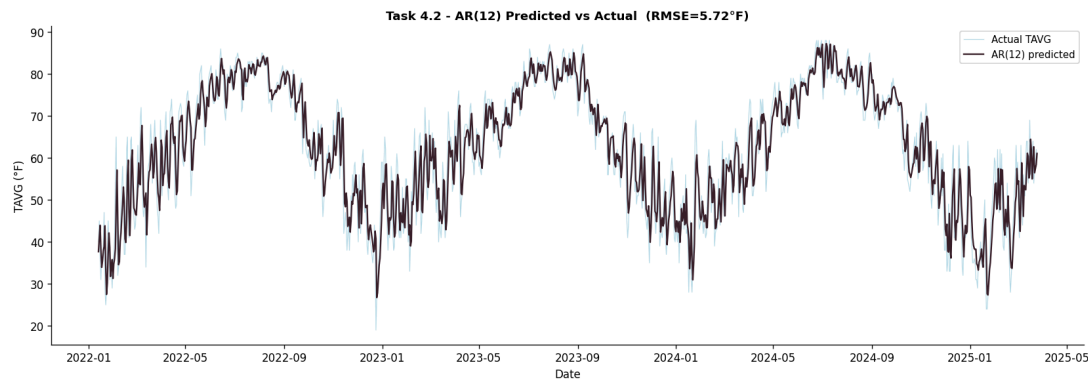
```
=====
Dep. Variable:          y      No. Observations:      1180
Model:                AutoReg(12)  Log Likelihood      -3585.027
Method:              Conditional MLE  S.D. of innovations      5.209
Date:                Sun, 12 Apr 2026  AIC              7198.055
Time:                23:30:39      BIC              7268.937
Sample:              12      HQIC              7224.791
                        1180
=====
```

```
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
const          0.0397      0.152        0.260      0.795      -0.259      0.338
y.L1           0.0052      0.029        0.178      0.859      -0.052      0.062
y.L2          -0.3865      0.029     -13.252      0.000      -0.444     -0.329
y.L3          -0.2040      0.031      -6.569      0.000      -0.265     -0.143
y.L4          -0.2375      0.032      -7.537      0.000      -0.299     -0.176
y.L5          -0.2291      0.032      -7.133      0.000      -0.292     -0.166
y.L6          -0.1605      0.032      -4.952      0.000      -0.224     -0.097
y.L7          -0.1543      0.032      -4.765      0.000      -0.218     -0.091
y.L8          -0.1035      0.032      -3.231      0.001      -0.166     -0.041
y.L9          -0.0920      0.031      -2.927      0.003      -0.154     -0.030
y.L10         -0.1172      0.031      -3.790      0.000      -0.178     -0.057
y.L11         -0.0407      0.029      -1.408      0.159      -0.097      0.016
y.L12         -0.0725      0.029      -2.509      0.012      -0.129     -0.016
=====
```

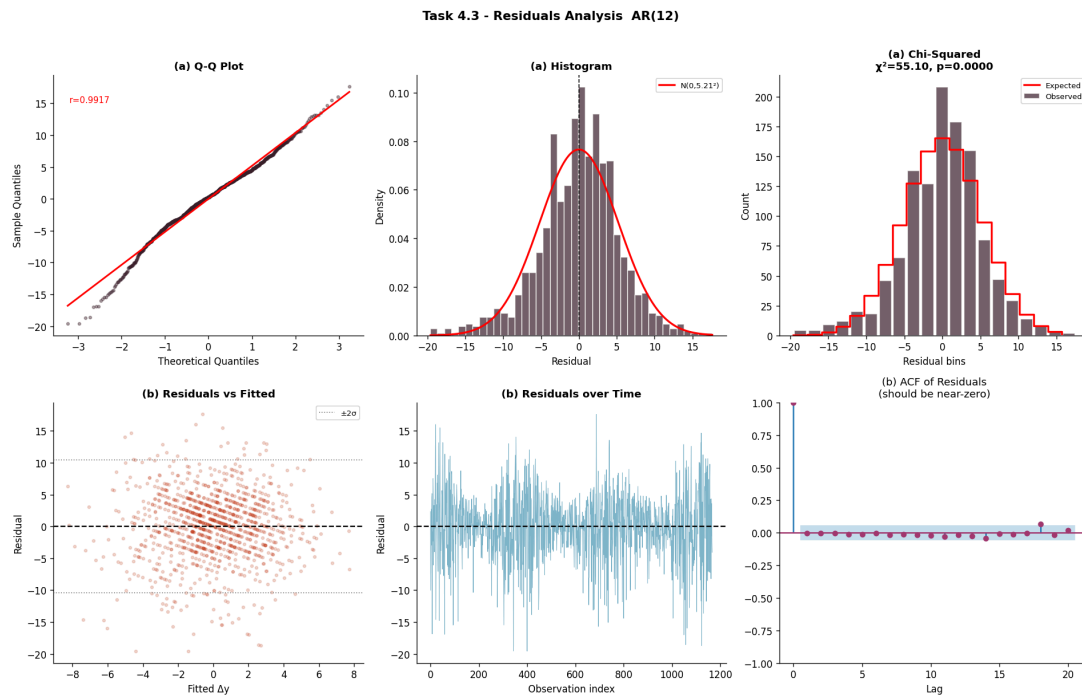
Roots

```
=====
              Real      Imaginary      Modulus      Frequency
-----
AR.1          1.0447      -0.5275j      1.1703      -0.0744
AR.2          1.0447      +0.5275j      1.1703      0.0744
AR.3          0.7472      -0.9222j      1.1869     -0.1416
AR.4          0.7472      +0.9222j      1.1869      0.1416
AR.5         -1.2176      -0.3462j      1.2659     -0.4559
AR.6         -1.2176      +0.3462j      1.2659      0.4559
AR.7         -0.8121      -0.9652j      1.2613     -0.3613
AR.8         -0.8121      +0.9652j      1.2613      0.3613
AR.9          0.2414      -1.2602j      1.2832     -0.2199
AR.10         0.2414      +1.2602j      1.2832      0.2199
AR.11        -0.2842      -1.2734j      1.3047     -0.2850
AR.12        -0.2842      +1.2734j      1.3047      0.2850
=====
```

AR(12) Training RMSE (original scale) = 5.7198°F



Task - 4.3 ->



Residuals mean = 0.000000

Residuals std = 5.2092°F

Chi-squared: $\chi^2=55.0977$, $p=0.0000$

-> Residuals approximately normal but not strictly normal by chi-squared.

With large n the test is highly sensitive. Q-Q plot is the primary assessment

Task - 4.4 ->

Although the lag-1 coefficient ($y.L1$) is not statistically significant, several higher-order lags (e.g., lags 2–10 and 12) are significant. This indicates that the predictive structure of the series is distributed across multiple past observations rather than being dominated by a single lag. Such behavior is common in seasonal time-series data where dependencies extend beyond immediate previous values.

- **Order selection:** PACF measures the direct relationship between $y(t)$ and $y(t-k)$ after removing intermediate lag effects. AR order p is set to the last lag where PACF exceeds $\pm 1.96/\sqrt{n}$. Beyond this, PACF falls within the confidence band indicating no additional predictive value.
- **Model fit:** The AR(p) model on the first-differenced series captures the short-term autocorrelation structure of daily temperature changes. Reconstructed back to the original scale it closely tracks the actual series. It is the most statistically principled of the three models.
- **Residuals:** The ACF of residuals should show near-zero values at all lags (white noise). If so, the AR model has fully captured the autocorrelation. The Q-Q plot and chi-squared test assess normality - minor tail deviations are expected with real weather data.

Task 5: Comparison of all the models (using the testing set)

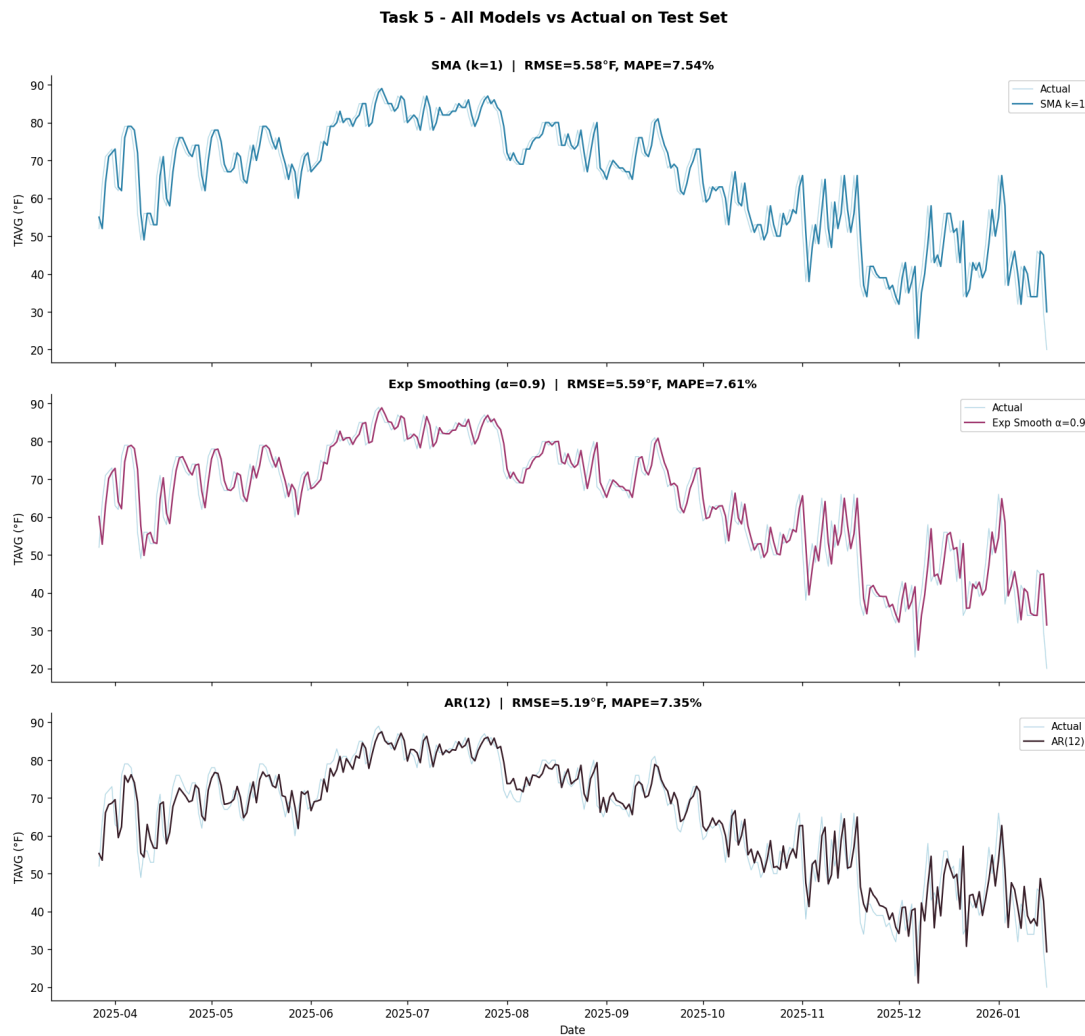
Models derived from training set:

SMA best k = 1 (Train RMSE = 5.8021°F)
 ES best α = 0.9 (Train RMSE = 5.8673°F)
 AR order p = 12 (Train RMSE = 5.7198°F)

== TEST SET PERFORMANCE ==

Model	Test RMSE	Test MAPE	Train RMSE
SMA ($k=1$)	5.5778	7.5355	5.8021
Exp Smooth ($\alpha=0.9$)	5.5927	7.6089	5.8673
AR(12)	5.1874	7.3476	5.7198

-> Best model on test set: AR(12) (lowest RMSE)



All three forecasting models were first derived using the training set and then evaluated on the final 20% test set using the same rolling one-step-ahead forecasting setup. This makes the comparison fair because each model predicts the next value using only past observed data and then updates after the actual value becomes available.

- Simple Moving Average (SMA): This model predicts temperature using the average of the most recent k observations. It is easy to interpret and provides a stable baseline, but because it gives equal weight to all points in the window, it tends to lag behind rapid transitions in temperature.
- Exponential Smoothing (ES): This model assigns higher weight to recent observations and lower weight to older ones. It reacts faster than SMA to local changes in the series and is often a stronger short-term forecasting baseline for highly autocorrelated data such as daily temperature.

-
- **AR(p) Model:** This model is the most statistically structured of the three. It uses PACF to select the lag order and models the differenced stationary series, then converts predictions back to the original temperature scale. Because daily temperature has strong temporal dependence, AR forecasting is theoretically well motivated.
 - **Best model:** The best model is selected empirically based on the lowest test RMSE and MAPE under a consistent rolling forecasting setup. Although AR models are theoretically suitable due to strong temporal autocorrelation in temperature data, actual performance depends on how well each model captures short-term dynamics in the test set.

From an IoT perspective, the results show how well future temperature can be predicted using only past sensor readings. This is useful for deciding how much historical data an edge device should retain and which forecasting method gives the best balance of simplicity and predictive accuracy.

Overall Conclusion =>

This project analyzed daily temperature data from NOAA using multiple time-series forecasting techniques, including Simple Moving Average, Exponential Smoothing, and AR(p) models. The analysis began by identifying strong non-stationary characteristics in the data, including annual seasonality, changing variance, and a cyclic mean pattern. After applying appropriate transformations such as differencing, the series was made suitable for time-series modeling.

Among the forecasting methods, Simple Moving Average provided a stable baseline but suffered from lag during seasonal transitions. Exponential Smoothing improved responsiveness by assigning greater weight to recent observations, making it more effective for short-term prediction. The AR(p) model, built using PACF-based lag selection and applied on a stationary differenced series, captured the temporal dependency structure more explicitly.

When evaluated on the test set using a consistent rolling one-step-ahead forecasting approach, the AR(12) model achieved the lowest RMSE and MAPE, indicating the best predictive performance among the three models. However, the performance differences were relatively small, suggesting that all three approaches are reasonable for forecasting this dataset.

From an IoT perspective, the results demonstrate that temperature can be effectively predicted using only past sensor readings, reducing the need for complex multi-sensor setups. This has implications for designing efficient edge-based forecasting systems with limited computational and storage resources. Overall, this project highlights the importance of combining statistical analysis with time-series modeling to understand and predict real-world environmental data in a robust and interpretable manner.